

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.3658395879746201, median 0.33043576109718076, std: 0.21791440

Gyroscope error (imu0): mean 0.6313911167131884, median 0.5824236917486224, std: 0.3321514229

Accelerometer error (imu0): mean 1.0339682711477771, median 0.9361813152629271, std: 0.548004859

Residuals

Reprojection error (cam0) [px]: mean 0.3658395879746201, median 0.33043576109718076, std: 0.21791

Gyroscope error (imu0) [rad/s]: mean 0.001562613290730948, median 0.0014414250968570403, std: 0.0

Accelerometer error (imu0) [m/s^2]: mean 0.0424053066114988, median 0.0383948490736683, std: 0.022

Transformation (cam0):

T_ci: (imu0 to cam0):

[[-0.69512001 0.71866341 -0.01819572 0.0139973]

[0.04606881 0.06978975 0.99649739 -0.04550621]

[0.71741609 0.69184702 -0.08162022 -0.10017374]

[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[-0.69512001 0.04606881 0.71741609 0.08369247]

[0.71866341 0.06978975 0.69184702 0.06242142]

[-0.01819572 0.99649739 -0.08162022 0.03742531]

[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)

-0.01806249587216875

Gravity vector in target coords: [m/s^2]

[-0.06305347 -9.7605658 -0.94646837]

Calibration configuration

Camera model: omni
Focal length: [1695.1353162186717, 1695.2777025334142]
Principal point: [738.3586774961138, 561.0063514994747]
Omni xi: 2.1719671881107505
Distortion model: radtan
Distortion coefficients: [-0.12749960535709404, 0.681843808523021, 0.0022260920512929593, 0.002702
Type: aprilgrid
Tags:
 Rows: 6
 Cols: 6
 Size: 0.055 [m]
 Spacing 0.0165 [m]

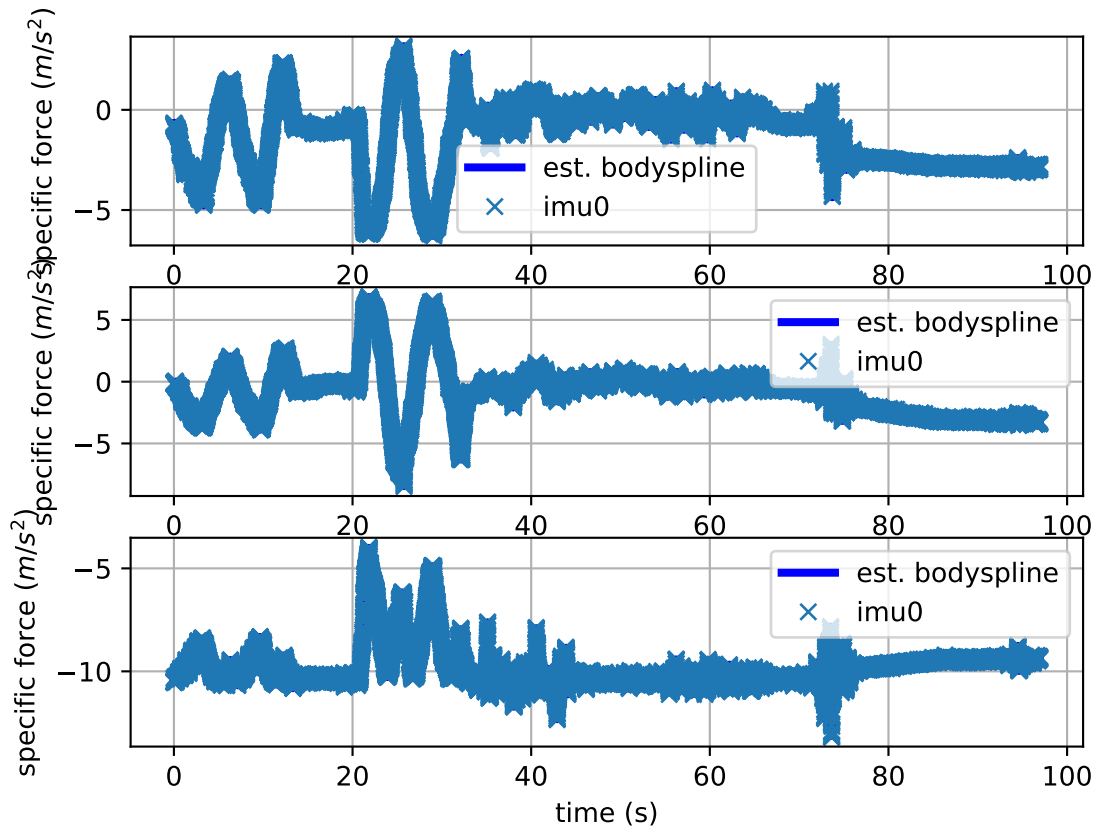
IMU configuration

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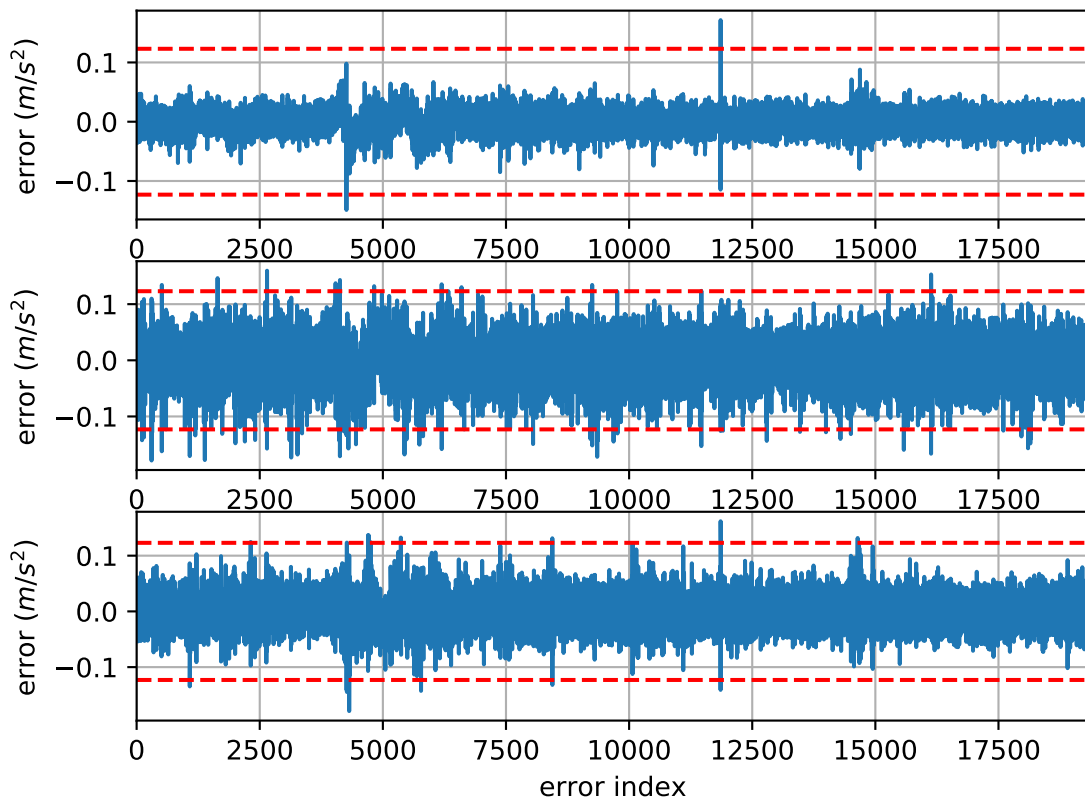
IMU0:

Model: calibrated
Update rate: 200.0
Accelerometer:
 Noise density: 0.0029
 Noise density (discrete): 0.04101219330881976
 Random walk: 0.00086
Gyroscope:
 Noise density: 0.000175
 Noise density (discrete): 0.0024748737341529162
 Random walk: 2.2e-05
T_i_b
 [[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

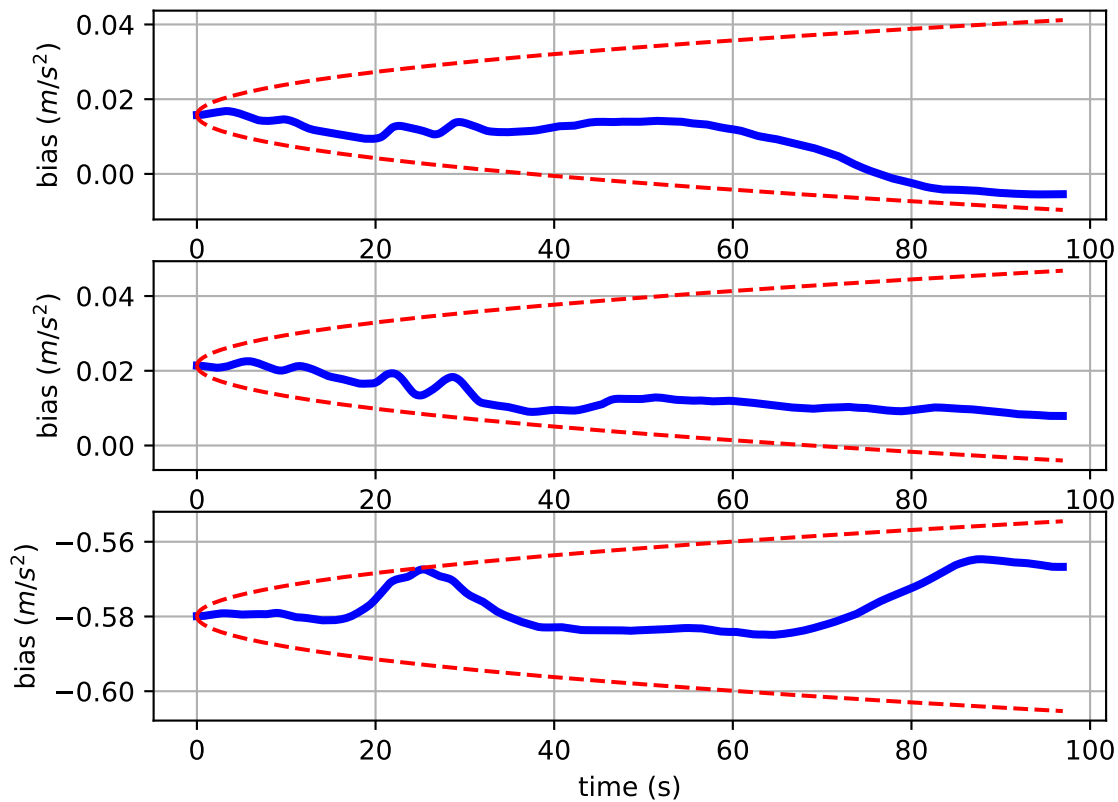
Comparison of predicted and measured specific force (imu0 frame)



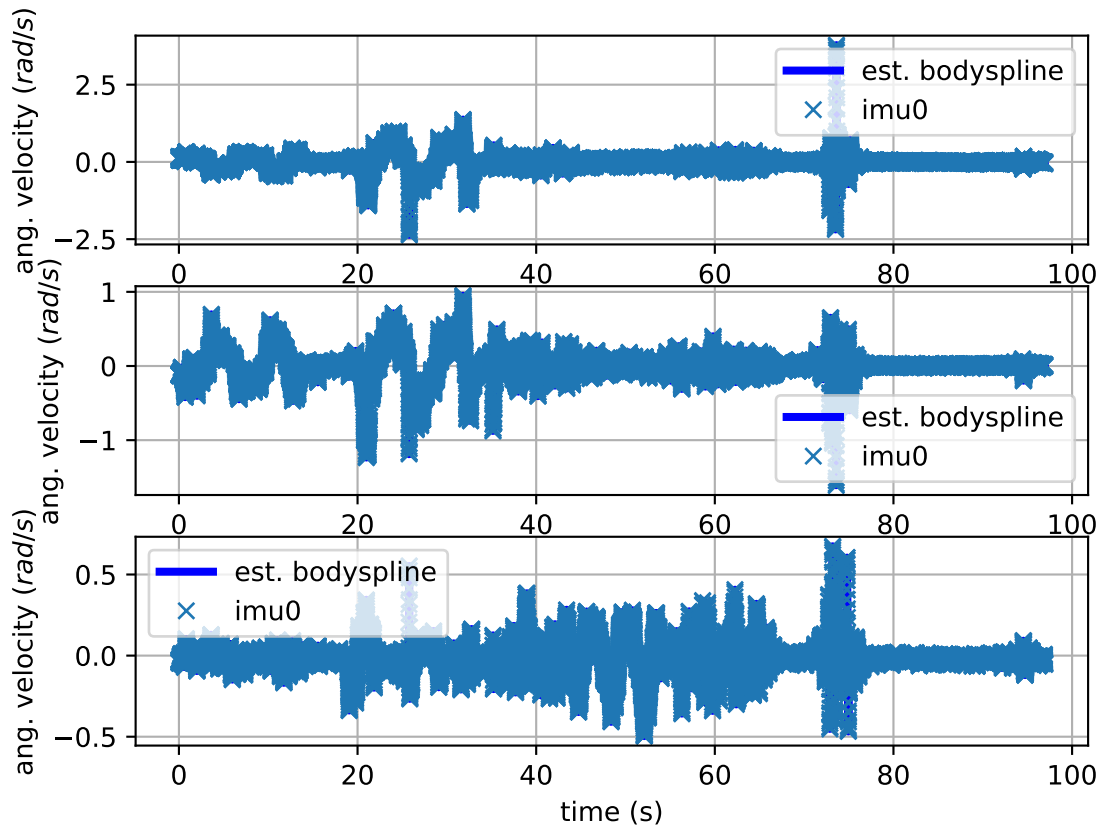
imu0: acceleration error



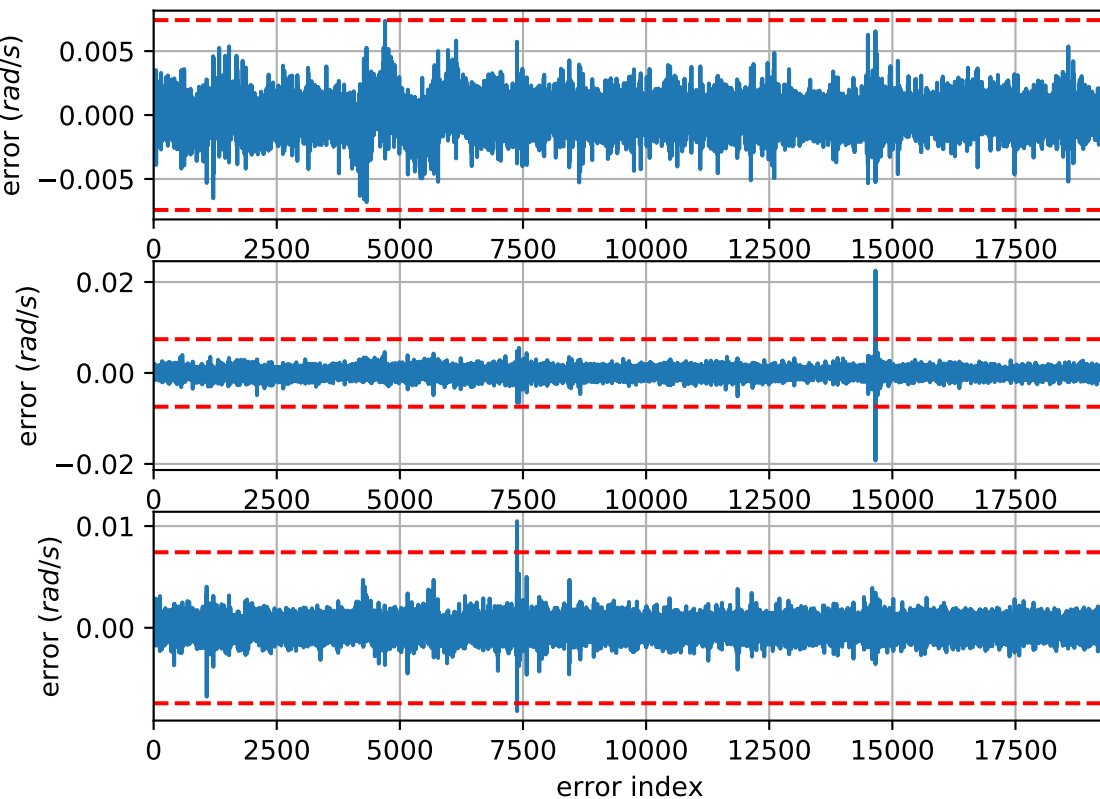
imu0: estimated accelerometer bias (imu frame)



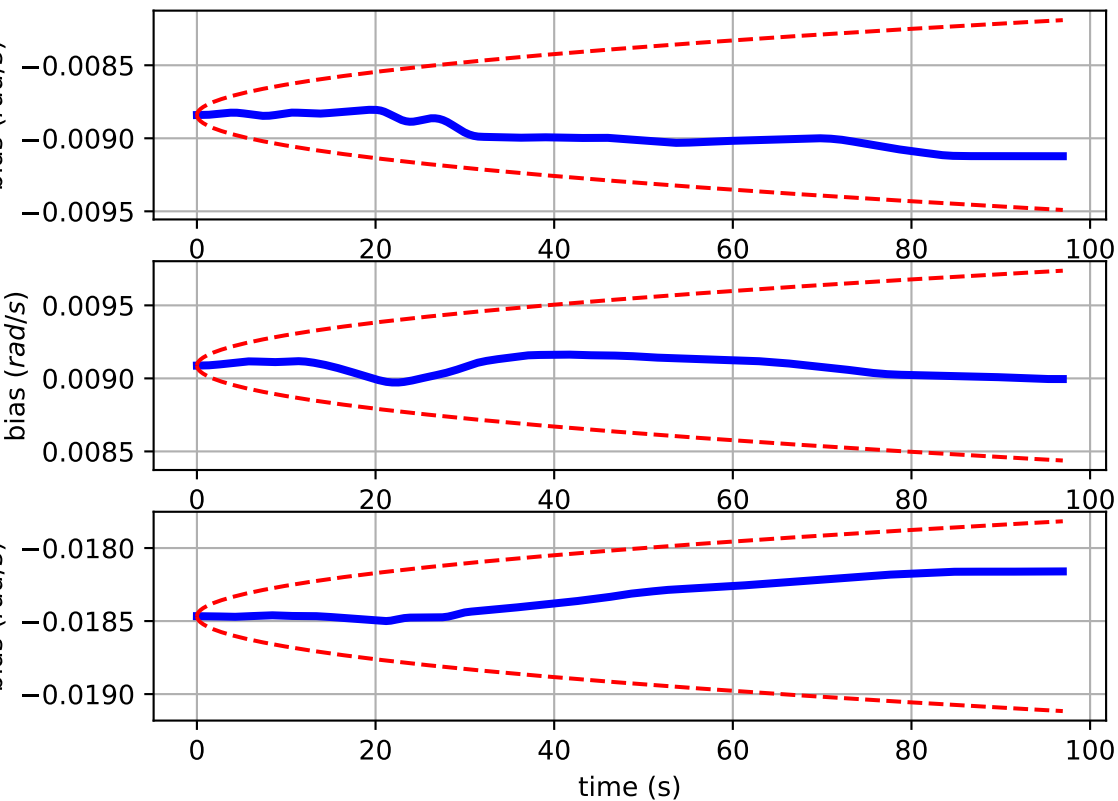
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

